

Pedro Miguel Domingos Marques

Software Developer

Lisbon, Portugal | pedrom02.dev@gmail.com | (+351) 939 915 944

LinkedIn: linkedin.com/in/pedro-marques-dev02 | GitHub: github.com/Pedrom2002 | Portfolio: portfolio-pedromarques-dev.vercel.app

SUMMARY

Full-stack Software Developer with a growing focus on AI and Data, shipping production systems across web, mobile, and desktop — from a regional energy-forecasting ML system to an AI-powered mobile app to a multi-tenant SaaS in production. Strong focus on testing, clean code, and automated deployment.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C#, Rust, Go

ML & Data: Scikit-learn, XGBoost, CatBoost, LightGBM, Optuna, Pandas, NumPy, SHAP, Feature Engineering

Backend: FastAPI, Flask, REST APIs, WebSockets, PostgreSQL, Redis

Frontend: React, React Native, Next.js, Expo, Tailwind CSS

DevOps & Cloud: Docker, Kubernetes, GitHub Actions, DVC, Pytest, AWS ECS, Azure, GCP Cloud Run, HuggingFace Spaces

EXPERIENCE

Software Development Intern | Apr 2026 – Sep 2026

QUIC Portugal, Lisbon

Core developer in a 2-person team responsible for all software and automation for the company's audiovisual event productions.

- Co-built and shipped Quic Platform to production, plus its companion React Native (Expo) app — from architecture to store-ready deployment.
- Delivered 3 client event systems end-to-end: guest management and accreditation for Quic Festival, the public GoalFest site (goalfest.pt) with SEO and a 3D venue model, and a real-time prize-draw system for ActivoBank.
- Owned cross-cutting quality and security: automated testing with coverage gates, dependency audits, and fixes to real access-control and file-validation vulnerabilities.

PROJECTS

Energy Forecast PT — Regional Energy-Consumption Forecasting (ML)

github.com/Pedrom2002/energy-forecast-pt

- Full-stack ML system forecasting hourly energy consumption across 5 Portuguese regions; production model reaches 1.44% MAPE and 0.998 R2 — a 61% RMSE improvement over baseline — trained on 40,000+ hourly samples (78 features) with Optuna over 5-fold time-series cross-validation.
- Served through a FastAPI backend (7 routers, 90% conformal intervals) and a React 19 dashboard, with Prometheus monitoring and 760+ automated tests (85% coverage gate); deployed on HuggingFace Spaces.

Stack: Python, XGBoost, LightGBM, CatBoost, Optuna, scikit-learn, FastAPI, React 19, Docker, GitHub Actions

Duelt — AI-Powered Household Task App for Couples

github.com/Pedrom2002/duelt

- Cross-platform mobile app (solo, architecture to deployment) with 13+ AI features via Gemini 2.5 Flash and a burnout-detection engine over 8 behavioural signals; client-side AES-256-GCM encryption with Argon2id KDF.
- 2,257 automated tests across 118 suites (95%+ coverage) and 37 Maestro E2E flows; CI/CD with EAS, CodeQL, and Dependabot.

Stack: React Native 0.83, Expo SDK 55, TypeScript, Firebase (Firestore / FCM / Vertex AI), Gemini 2.5 Flash, Jest, Maestro

Quic Platform — Multi-Tenant Event & Client-Communication SaaS

github.com/Pedrom2002/quic-platform

- Multi-tenant SaaS for audiovisual event productions (events, ticketing, inventory, investor relations), in production for QUIC with a companion React Native app; 15+ PostgreSQL tables with row-level security across 12+ API domains.
- Stripe ticketing with signed webhooks and idempotent processing; 8 Gemini-powered AI endpoints, a serverless notification pipeline (QStash + Vercel Cron), and a Golden Circle investor area with KYC and hardened access control.

Stack: Next.js 16, React 19, TypeScript, Supabase (PostgreSQL / RLS / Realtime), Stripe, Upstash QStash, Expo / React Native, Vitest, Playwright

CERTIFICATIONS (COURSERA)

Data Science Professional Certificate (IBM) • Advanced Data Analytics Professional Certificate (Google) • AI Engineering Professional Certificate (IBM) • Advanced Statistics for Data Science (Johns Hopkins) • Data Visualization with Tableau (UC Davis)

EDUCATION

BSc in Computer Engineering — ISTEC Lisboa, Portugal (post-laboral) | 2026 – 2028

CTeSP in Mobile Devices Development — ISTEC Lisboa, Portugal (EQF Level 5) | 2024 – 2026

Languages: Portuguese (Native), English (C1), Spanish (B1)